MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2025
CHAPTER 2: SEQUENCES (PART 2)

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Goal of this section:

(1) Subsequences
(2) Bolzano-Weierstrass Theorem
(3) Cauchy sequences

Reference books:

- Robert G. Bartle, Donald R. Sherbert: Introduction to Real Analysis (Wiley, 2011, Fourth Edition)
- Kenneth A. Ross: Elementary Analysis: The Theory of Calculus (Springer, 2013, Second Edition)

1. Subsequences

1.1. Subsequences.

Definition 1.1. Let (x_n) be a sequence in $\mathbb{R}$ and let $n_1 < n_2 < \dots$ be a strictly increasing sequence of numbers in $\mathbb{N}$. Then the sequence $(x_{n_1}, x_{n_2}, x_{n_3}, \dots)$ is called a subsequence of (x_n).

Example 1.2. (1) Let $x_n = n^2$. A subsequence is $(3^2, 6^2, 9^2, 12^2, \dots)$, which takes the even terms in (x_n).
(2) Let $x_n = (-1)^n$. A subsequence is x_{2n} is a constant sequence with all terms equal to 1.

Example 1.3. (1) Let

$$x_n = \begin{cases} \frac{1}{n^2+1} & n \text{ is odd} \\ \frac{-1}{n+1} & n \text{ is even} \end{cases}$$

Then (x_{2k}) and (x_{2k-1}) are two subsequences of (x_n). Show that the two subsequences are convergent.
(2) Can we conclude that (x_n) is convergent?

Exercise 1.4. Let (x_n) be a bounded sequence. Then any subsequence of (x_n) is also bounded.

Exercise 1.5. Let (y_n) be a subsequence of (x_n) and let (z_n) be a subsequence of (y_n). Use $\epsilon - N$ definition to show (z_n) is also a subsequence of (x_n).

1.2. **Convergence for subsequences.**

Theorem 1.6. *(Convergence of subsequence of a convergent sequence) Let (x_n) be a sequence in $\mathbb{R}$ converging to x, then any subsequence of (x_n) also converges to x.*

Proof. Let $\epsilon > 0$. Since (x_n) is a convergent sequence, there exists $N(\epsilon) > 0$ such that $|x_n - x| < \epsilon$ for any $n > N(\epsilon)$. Now for any $k > N(\epsilon)$, $n_k \geq k > N(\epsilon)$, we also have:

$$|x_{n_k} - x| < \epsilon.$$

By definition of convergence, (x_{n_k}) is also convergent. □

Example 1.7. Let $a > 1$. Find $\lim\ a^{1/n}$. Let $x_n = a^{1/n}$. (We used squeeze theorem to find this before and here we use another method.) One first uses Monotone Convergence Theorem to show the limit exists. Let $x = \lim\ x_n$. We consider the subsequence (x_{2n}) of (x_n). Then

$$x_{2n}^2 = x_n.$$

By Theorem 1.6, $\lim\ x_{2n} = x$ and $\lim\ x_{2n}^2 = x^2$. Hence, $x^2 = x$ and so $x = 0$ or 1. But $x_n > 1$ and so $x \geq 1$. Thus, $x = 1$.

Exercise 1.8. Does the converse of Theorem 1.6 hold?

Exercise 1.9. Let (x_n) be a monotone sequence. Suppose (x_n) has a convergent subsequence. Determine if the statement is true: (x_n) is also convergent.

1.3. **Divergence Criteria.**

Proposition 1.10. *(Divergence by finding two subsequences converging to different limits) If (x_n) has two subsequences converging to two different limits, then (x_n) is divergent.*

Proof. This follows from Theorem 1.6. □

Example 1.11. Determine whether the sequence $(2^n + (-2)^n)$ converges.

Example 1.12. Let $x_n \geq 0$ for all n. Suppose (x_n) is convergent. Determine when $\lim\ (-1)^n x_n$ exists.

2. Bolzano-Weiterstrass Theorem

We have seen two main examples of divergent sequences:

(1) (n)
(2) $((-1)^n)$

The first sequence is not bounded, while the second one is bounded. In the second one, it is easy to observe that $((-1)^n)$ has convergent subsequences $((-1)^{2n})$ and $((-1)^{2n+1})$. We shall now prove that this is a more general phenomenon:

2.1. Monotone Subsequence Theorem.

Theorem 2.1. *(Monotone Subsequence Theorem) Let (x_n) be a sequence in $\mathbb{R}$. Then the sequence has a monotone subsequence.*

Proof. We shall first introduce a terminology. We say that x_m is a peak in the sequence (x_n) if $x_m \geq x_n$ for all $n \geq m$. We now consider two possibilities:

Case 1: (x_n) has infinitely many peaks.
Then we can find $m_1 \leq m_2 \leq \ldots$ such that $x_{m_1} \geq x_{m_2} \geq \ldots$ are the peaks. By definition, $x_{m_1}, x_{m_2}, \ldots$, is a monotone subsequence of (x_n).

Case 2: (x_n) does not have infinitely many peaks.
Let $x_{m_1}, \ldots, x_{m_k}$ be *all* those finitely many peaks, with $m_1 < m_2 < \ldots < m_k$. We then pick $s_1 = m_k + 1$. Since s_1 is not a peak, we can find $s_2 > s_1$ such that $x_{s_2} > x_{s_1}$. Again s_2 is not a peak, we can find $s_3 > s_2$ such that $x_{s_3} > x_{s_2}$. One repeats the same process to find $x_{s_4}, x_{s_5}, \ldots$. Then $x_{s_1} < x_{s_2} < x_{s_3} < \ldots$ is a monotone sequence. □

Exercise 2.2. Describe the peaks of the following sequences:

(a) $x_n = 1 + \frac{(-1)^n}{n}$;
(b) $x_n = (-1)^n(1 - \frac{1}{n})$.

Are there infinitely many or finitely many peaks?

Exercise 2.3. Find a monotone subsequence of the following examples:

(1) $(2^n + (-1)^n 2^n)$; and
(2) $(\sin\left(\frac{n\pi}{6}\right))$.

2.2. The Bolzano-Weierstrass Theorem.

Theorem 2.4. *(Bolzano-Weierstrass Theorem) A bounded sequence of real numbers has a convergent subsequence.*

Proof. Let (x_n) be a bounded sequence. By Theorem 2.1, there exists a monotone subsequence. Then such monotone subsequence must also be bounded (Exercise 1.4). Now, by Monotone Convergence Theorem , the subsequence is convergent. □

Alternate proof.
Step 1: Construct nested closed intervals Let (x_n) be a bounded sequence. Then there exists a closed interval I_1 such that $x_n \in I_1$ for all $n \in \mathbb{N}$. Bisect I_1 into two closed intervals I_1' and I_2' with the same length. Let

$$A_1' = \left\{n \in \mathbb{N} : x_n \in I_1'\right\}, \quad B_1' = \left\{n \in \mathbb{N} : x_n \in I_1''\right\}$$

If A_1' is infinite, let $I_2 = I_1'$. If A_1' is finite, then B_1' is infinite, and let $I_2 = I_1''$.

We now again bisect I_2' into two equal closed intervals A_2' and B_2'. Similarly, if A_2' is infinite, let $I_3 = A_2'$. If A_2' is finite, then B_2' is infinite and let $I_3 = B_2'$. We repeat the process to obtain I_n, which forms a nested closed intervals. Hence, by a result in last chapter, we have that $\cap_n I_n \neq \emptyset$. Let $x \in \cap_n I_n$.
Step 2: Form a subsequence

Now, we now pick a subsequence x_{n_k} such that $x_{n_k} \in I_k$. It remains to show that (x_{n_k}) converges to x. Note that x_{n_k}, x are in I_k and so

$$|x_{n_k} - x| < b_k - a_k = \frac{1}{2^{k-1}}(b_1 - a_1).$$

Since $\lim \frac{1}{2^{k-1}}(b_1 - a_1) = 0$, we have $\lim x_{n_k} = x$. □

The Bolzano-Weierstrass Theorem shows the existence of a convergent subsequence of a bounded sequence. As shown in the proof, that is to find monotone subsequences.

Example 2.5. Let (x_n) be a sequence of the form:

$$1, \frac{1}{2}, 1, \frac{2}{3}, \frac{1}{3}, 1, \frac{3}{4}, \frac{2}{4}, \frac{1}{4}, 1, \frac{4}{5}, \ldots$$

How many limits of convergent subsequences of (x_n) that you can find?

3. Cauchy sequences

3.1. Cauchy sequences.

Definition 3.1. A sequence (x_n) in $\mathbb{R}$ is said to be Cauchy if for any given $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for any $n, m > N$,

$$|x_n - x_m| < \epsilon.$$

Exercise 3.2. Draw a real line to put ϵ and $x_{N+1}, x_{N+2}, \ldots$ to understand the meaning of the definition.

Example 3.3. (1) The sequence $(\frac{1}{n})$ is a Cauchy sequence.

Proof: Let $\epsilon > 0$. We find an integer $N > \frac{2}{\epsilon}$. Then, for any $n, m > N$,

$$\left|\frac{1}{n} - \frac{1}{m}\right| < \left|\frac{1}{n}\right| + \left|\frac{1}{m}\right| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

(2) (Non-example) The sequence $(n + \frac{(-1)^n}{n})$ is not a Cuachy sequence.

Proof: Let $\epsilon = 1$. For any integer $N \in \mathbb{N}$, we can choose $n = N + 1$ and $m = N + 5$,

$$\left|\left(N + 5 + \frac{(-1)^{N+5}}{N+5}\right) - \left(N + 1 + \frac{(-1)^{N+1}}{N+1}\right)\right| \geq |4| - \frac{1}{N+5} - \frac{1}{N+1} \geq 2 > \epsilon.$$

We have many examples from convergent sequences:

Theorem 3.4. *Any convergent sequences in $\mathbb{R}$ is Cauchy.*

Proof. Step 1: Inequality from convergent sequences
Let (x_n) be a convergent sequence. Let $x = \lim x_n$. For $\epsilon > 0$, let $N(\epsilon) \in \mathbb{N}$ such that for any $n > N(\epsilon)$,

$$|x_n - x| < \epsilon.$$

Step 2: Bound the distance for large n, m
Let $\epsilon_0 > 0$. Then for any $n, m > N(\frac{\epsilon_0}{2})$,

$$|x_n - x_m| = |x_n - x + x - x_m| \leq |x_n - x| + |x - x_m| < \frac{\epsilon_0}{2} + \frac{\epsilon_0}{2}.$$

Hence, (x_n) forms a Cauchy sequence. □

3.2. **Cauchy sequence is convergent.**

Lemma 3.5. *Any Cauchy sequence in $\mathbb{R}$ is bounded.*

Proof. Let (x_n) be a Cauchy sequence. Let $\epsilon > 0$. Let $N \in \mathbb{N}$ such that for any $n, m > N$,

$$|x_n - x_m| < \epsilon.$$

In particular, for any $n > N$, $|x_n - x_{N+1}| < \epsilon$. Thus

$$x_{N+1} - \epsilon < x_n < x_{N+1} + \epsilon$$

Hence, $|x_n| < |x_{N+1}| + \epsilon$. Thus, (x_n) is bounded by

$$\max\{|x_1|, \ldots, |x_N|, |x_{N+1}| + \epsilon\}$$

□

3.3. **Convergence of Cauchy sequence.**

Theorem 3.6. *Any Cauchy sequence is convergent.*

Proof. Step 1: Find the expected limit by using BW theorem.
Lemma 3.5 implies that the Cauchy sequence is bounded. By the Bolzano-Weierstrass theorem, there exists a convergent subsequence (x_{n_k}). Let $x^* = \lim_k x_{n_k}$.

Step 2: Obtains bounds from Cauchy sequences and convergence of subsequence.
We shall now prove that (x_n) converges to x^*. Fix an $\epsilon > 0$. Then, by the definition of Cauchy sequence, we has $N \in \mathbb{N}$ such that for $n, m > N$,

$$|x_n - x_m| < \frac{\epsilon}{2}$$

By the definition of convergence, we also have $N' \in \mathbb{N}$ such that for $n_k > N'$,

$$|x_{n_k} - x^*| < \frac{\epsilon}{2}.$$

Step 3: Show convergence.
Thus, for any $n > \max\{N, N'\}$, we find some $n_{k'} > \max\{N, N'\}$ (why $n_{k'}$ exists) such that

$$|x_n - x^*| = |x_n - x_{n_{k'}} + x_{n_{k'}} - x^*| \leq |x_n - x_{n_{k'}}| + |x_{n_{k'}} - x^*| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

□

3.4. **Examples.** Using Cauchy sequences to show convergence is sometimes easier since sometimes we may not know the limit in piori, and also Monotone Convergence Theorem may not apply.

Example 3.7. Let

$$y_n = \frac{1}{1!} - \frac{1}{2!} + \ldots + \frac{(-1)^{n+1}}{n!}.$$

Then, for $m > n \geq 1$,

$$\begin{aligned} |y_m - y_n| &= |\frac{(-1)^{n+2}}{(n+1)!} + \ldots + \frac{(-1)^{m+1}}{m!}| \\ &\leq \frac{1}{(n+1)!} + \ldots + \frac{1}{m!} \\ &\leq \frac{1}{2^n} + \ldots + \frac{1}{2^{m-1}} \\ &\leq \frac{1}{2^{n-1}} \end{aligned}$$

Since $\lim \frac{1}{2^{n-1}} = 0$, we have $|y_m - y_n| < \epsilon$ for large n. This shows that (y_n) is a Cauchy sequence.

Exercise 3.8. Do you know (without a proof) the limit of Example 3.7?

Exercise 3.9. A sequence (x_n) is said to be contractive if there exists a number C such that $0 < C < 1$ and for all $n \in \mathbb{N}$,

$$|x_{n+2} - x_{n+1}| \leq C|x_{n+1} - x_n|.$$

Prove that any contractive sequence is Cauchy.

Proof: Let $m > n$. Then

$$\begin{aligned}
|x_m - x_n| &= |x_m - x_{m-1} + x_{m-1} - \ldots + x_{n+1} - x_n| \\
&\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \ldots + |x_{n+1} - x_n| \\
&\leq (C^{m-2} + C^{m-3} + \ldots + C^{n-1})|x_2 - x_1| \\
&= \frac{C^{n-1} - C^{m-1}}{1 - C}|x_2 - x_1| \\
&\leq \frac{C^{n-1}}{1 - C}|x_2 - x_1|
\end{aligned}$$

Let $\epsilon > 0$. Since $\lim_{n\to\infty} \frac{C^{n-1}}{1-C}|x_2 - x_1| = 0$, we can choose large N, such that for any $n, m > N$, $|x_m - x_n| \leq \frac{C^{n-1}}{1-C}|x_2 - x_1| < \epsilon$. This shows (x_n) is a Cauchy sequence.

Exercise 3.10. Recall the Fibonacci sequence (f_n) is defined as: $f_1 = f_2 = 1, f_{n+2} = f_{n+1} + f_n$. Let $x_n = \frac{f_n}{f_{n+1}}$. Show that (x_n) is contractive.